

CAPSTONE PROJECT

Customer Churn Prediction Using Machine Learning

Leveraging Predictive Models to Proactively Drive Customer Retention

Prepared by:

Valentine Amoso

 Data Analyst

 Mathematics Teacher

 AI in Business Intelligence Learner

 May 2026

01 / THE CHURN CHALLENGE

⚠ Business Problem

Customer churn is a critical and costly business challenge for telecom providers.

- 📉 **Revenue Drainage:** Customer loss leads to immediate subscription revenue declines.
- 📁 **High Acquisition Costs:** Winning a new customer is up to 5x more expensive than retaining one.
- Stifled Business Growth:** High attrition rates drag overall enterprise scalability.

🎯 Capstone Project Goal

Develop a high-precision ML framework to predict churn before it happens.

- 🔧 **Proactive Intervention:** Flag vulnerable customers automatically inside the pipeline.
- 👤 **Targeted Retentions:** Empower account managers with precise churn indicators.
- 📈 **Save Customer Equity:** Maintain steady growth and maximize lifetime value (LTV).

02 / DATASET OVERVIEW

IBM Telco Churn Dataset

A clean, standard analytical cohort used to construct predictive indicators from customer records.

-  **7,043 Customer Records:** Broad historic demographic data.
-  **21 Original Features:** Mix of numeric and categorical identifiers.
-  **Target Feature (Churn):** Binomial category representing whether the customer departed or stayed.

Category	Monitored Elements
Demographics	Gender, Seniority, Partner, Dependents
Account info	Tenure, Contract Type, Payment Method
Subscriptions	Phone, Internet, Security, Tech Support
Billing Details	Monthly Charges, Total Charges

03 / EXPLORATORY DATA ANALYSIS

Key Analytical Revelations

Data exploration uncovered structured patterns separating active accounts from churning cohorts.

-  **Distribution Balance:** While the majority remain retained, churn makes up a significant minority.
-  **Month-to-Month High Attrition:** Customers on rolling month contracts churn far more frequently.
-  **High Billing Sensitivity:** Elevated Monthly Charges are positively correlated with churn events.

600 × 400

04 / PREPROCESSING PIPELINE

Preparing Data for ML Models

To ensure accurate predictive behavior, raw inputs underwent rigorous cleaning to standardize scales and eliminate noise.

- ✓ **Numeric Conversion:** Coerced TotalCharges columns to float metrics, handling spaces.
- ✓ **Missing Value Management:** Safely removed clean null boundaries without biasing metrics.
- ✓ **Unneeded Identifier Exclusions:** Dropped unique customerID columns.
- ✓ **Label & Encoding:** Converted multi-class nominal features into numeric maps.
- ✓ **Standardization:** Scaled numeric inputs to uniform distributions.



Pipeline Result

Clean, scaled, and normalized arrays optimized to power sensitive estimators without risk of feature leakage.

05 / FEATURE ENGINEERING

1. Tenure Group Allocation

Transformed raw numeric months of tenure into structural cohorts to track retention lifecycle segments:

-  **New Accounts:** Initial setup phase risk points.
-  **Developing:** Adapting to subscription.
-  **Established:** Stabilized regular users.
-  **Loyal:** High customer lifetime value advocates.

2. Derived Value Metric

Constructed monthly revenue efficiency factors to measure exact spending pacing against total lifecycle values:

$$\text{Revenue}_{\text{PerMonth}} = \frac{\text{TotalCharges}}{\text{Tenure}}$$

Note: Handled edge boundaries where Tenure is zero to avoid undefined divides.

06 / MACHINE LEARNING MODELS



Logistic Regression

Constructed a standard probability-based linear baseline model. Provides maximum feature interpretability and simple coefficient sign evaluation.



Decision Tree

Introduced non-linear relationship segmenting. Good at capturing visual decision splits though vulnerable to overfitting without depth limits.



Random Forest

Deployed a powerful ensemble bagging framework. Aggregates several deep estimators to improve accuracy and stabilize random errors.

07 / MODEL PERFORMANCE

Evaluated Model	Classification Accuracy	ROC-AUC Score	Status / Outcome
Logistic Regression	79.53%	83.47%	 Selected Best Estimator SELECTED
Random Forest	78.96%	81.83%	Competitive baseline metrics
Decision Tree	71.71%	64.16%	Highly unstable and overfit

 **Why Logistic Regression succeeded:**
Not only did it maintain the highest classification correctness and trade-off ratios (ROC-AUC), but it also remains computationally lightweight, easily explainable, and produces direct coefficients for marketing teams.

08 / FEATURE IMPORTANCE



Top Model Predictors

The regression model highlighted which behaviors strongly dictate customer transitions.

-  **Spending Thresholds:** Higher overall charges have the highest predictive weight.
-  **Lifespan Tenure:** Active customer age acts as a protective shield against churn.

09 / STRATEGIC BUSINESS INSIGHTS

⬇️ What Drives Churn?

Critical behaviors indicating a high risk of imminent account cancellation:

- ⚠️ **Short Tenure:** New accounts drop out heavily in months 1–3.
- 📄 **Month-to-Month Contracts:** No long-term lock-in barriers.
- 💳 **High Rates:** Pricing complaints drive search for alternatives.
- 📄 **Electronic Checks:** Friction-prone manually processed payments.

⬆️ What Reduces Churn?

Protective behaviors that secure and extend standard lifetime limits:

- 🛡️ **Long-term Contracts:** One and two-year commitments have high retention.
- 👤 **Security Subscriptions:** Users with online security features stay longer.
- 🎧 **Tech Support:** Active customer service reduces client frustration.

10 / STRATEGIC RECOMMENDATIONS

- **1. Onboarding Enhancements:** Target fresh sign-ups with proactive check-ins to handle onboarding issues.
- **2. Promoted Contracts:** Build incentive plans (such as discounts) to switch rolling users to yearly commitments.
- **3. Price Adjustments:** Create customized bundle rates for price-sensitive high-risk classes.
- **4. Push Core Support:** Bundle Online Security and Tech Support features into baseline offerings.
- **5. Safe Payment Upgrades:** Encourage paperless and automated direct payments over electronic checks.

📈 Predicted Impact

By addressing these 5 high-priority churn avenues with automated system flags:

📈 Targeted retention efficiency is estimated to improve customer lifespan and lower overall attrition cost ratios.

Project Conclusion

Developed an end-to-end Machine Learning pipeline to successfully forecast customer churn.

79.53%

ACCURACY

83.47%

ROC-AUC

100%

INTERPRETABLE

Using **Logistic Regression** as the final selected framework, business intelligence units can flag at-risk accounts in real-time, implement preemptive retention steps, and secure steady corporate revenue growth.

Thank You | Valentine Amoso